

Week_6_GIT/Exercise_3_Branching_and_Merging

Step 1: Open Git Bash

Open Git Bash.

Step 2: Navigate to Your Repository

```
cd ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling
```

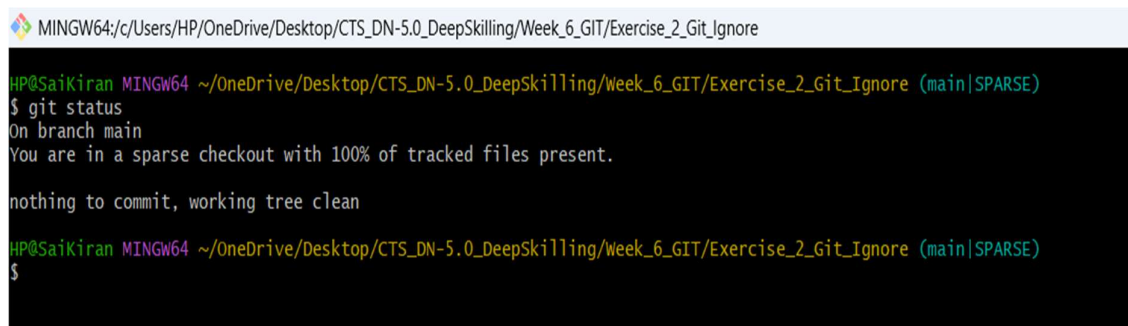
Verify you're inside the repository:

```
git status
```

Expected Output:

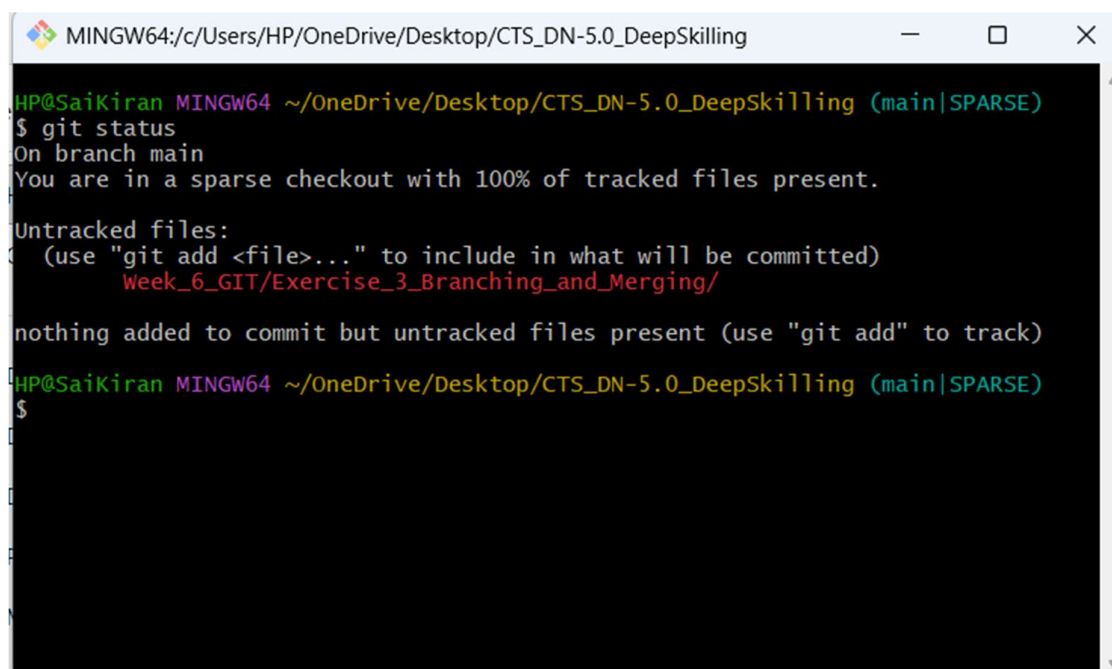
On branch main

nothing to commit, working tree clean



```
MINGW64/c:/Users/HP/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling/Week_6_GIT/Exercise_2_Git_Ignore
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling/Week_6_GIT/Exercise_2_Git_Ignore (main|SPARSE)
$ git status
On branch main
You are in a sparse checkout with 100% of tracked files present.

nothing to commit, working tree clean
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling/Week_6_GIT/Exercise_2_Git_Ignore (main|SPARSE)
$
```



```
MINGW64/c:/Users/HP/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling (main|SPARSE)
$ git status
On branch main
You are in a sparse checkout with 100% of tracked files present.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Week_6_GIT/Exercise_3_Branching_and_Merging/

nothing added to commit but untracked files present (use "git add" to track)
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling (main|SPARSE)
$
```

Part 1 – Branching

Step 3: Create a New Branch

Create a branch named **GitNewBranch**.

```
git branch GitNewBranch
```

Step 4: View All Branches

```
git branch
```

or

```
git branch -a
```

Expected Output:

```
GitNewBranch
```

```
* main
```

* indicates the currently active branch.

Step 5: Switch to the New Branch

```
git checkout GitNewBranch
```

Or using the newer command:

```
git switch GitNewBranch
```

Expected Output:

```
Switched to branch 'GitNewBranch'
```

Step 6: Verify Current Branch

```
git branch
```

Expected Output:

```
* GitNewBranch
```

```
main
```

```
HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git branch GitNewBranch

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git branch
  GitNewBranch
* main

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git branch -a
  GitNewBranch
* main
remotes/origin/HEAD -> origin/main
remotes/origin/main

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git switch GitNewBranch
D      branch.txt
Switched to branch 'GitNewBranch'

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ git branch
* GitNewBranch
  main

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ |
```

Step 7: Create a New File

Create a file:

```
echo "This file is created in GitNewBranch." > branch.txt
```

Verify:

```
cat branch.txt
```

Expected Output:

This file is created in GitNewBranch.

Step 8: Check Git Status

```
git status
```

Expected Output:

Untracked files:

```
branch.txt
```

Step 9: Stage the File

```
git add branch.txt
```

Step 10: Commit the Changes

```
git commit -m "Add branch.txt in GitNewBranch"
```

Expected Output:

```
[GitNewBranch xxxxxx]
```

Add branch.txt in GitNewBranch

Step 11: Verify Status

```
git status
```

Expected Output:

nothing to commit, working tree clean

This completes the **Branching** section.

```
MINGW64:/c:/Users/HP/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ echo "This file is created in GitNewBranch." > branch.txt

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ cat branch.txt
This file is created in GitNewBranch.

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ git status
on branch GitNewBranch
You are in a sparse checkout with 100% of tracked files present.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  ./

nothing added to commit but untracked files present (use "git add" to track)

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ git status -u
on branch GitNewBranch
You are in a sparse checkout with 100% of tracked files present.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  1_git_status_verified.png
  2_git_status_after_creating_this_folder.png
  3_Branching.png
  4_branch.txt_is_added_to_GitNewBranch.png
  5_git_diff.png
  6_Merging.png
  branch.txt

nothing added to commit but untracked files present (use "git add" to track)

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ git add branch.txt
warning: in the working copy of 'week_6_GIT/Exercise_3_Branching_and_Merging/branch.txt', LF will be replaced by CRLF the next time Git touches it

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ git commit -m "Add branch.txt in GitNewBranch in Ex3"
[GitNewBranch 8a2ea4a] Add branch.txt in GitNewBranch in Ex3
1 file changed, 1 insertion(+)
 create mode 100644 week_6_GIT/Exercise_3_Branching_and_Merging/branch.txt

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ git status
on branch GitNewBranch
You are in a sparse checkout with 100% of tracked files present.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  1_git_status_verified.png
  2_git_status_after_creating_this_folder.png
  3_Branching.png
  4_branch.txt_is_added_to_GitNewBranch.png
  5_git_diff.png
  6_Merging.png

nothing added to commit but untracked files present (use "git add" to track)

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ |
```

Part 2 – Merging

Step 12: Switch Back to Main Branch

git checkout main

or

git switch main

Expected Output:

Switched to branch 'main'

Step 13: Compare the Branches

View the differences between **main** and **GitNewBranch**.

git diff main GitNewBranch

Git displays the changes present in GitNewBranch but not yet in main.

This corresponds to listing the differences between the trunk and the branch.

```
MINGW64/c/Users/HP/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (GitNewBranch|SPARSE)
$ git switch main
Switched to branch 'main'

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git diff main GitNewBranch
diff --git a/branch.txt b/Week_6_GIT/Exercise_3_Branching_and_Merging/branch.txt
similarity index 100%
rename from branch.txt
rename to Week_6_GIT/Exercise_3_Branching_and_Merging/branch.txt

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ |
```

Step 15: Merge the Branch into Main

git merge GitNewBranch

Expected Output:

Updating xxxxxx..xxxxxx

Fast-forward

branch.txt | 1 +

or

Merge made by the 'ort' strategy.

depending on the commit history.

Step 16: View Commit History

`git log --oneline --graph --decorate`

Expected Output (example):

- * abc1234 (HEAD -> main) Add branch.txt in GitNewBranch
- * def5678 Previous Commit
- * ghi9012 Initial Commit

This command displays the commit graph after merging.

A screenshot of a terminal window with a black background and white text. The window title bar shows the path 'MINGW64:/c/Users/HP/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_...'. The terminal content shows a user 'HP@SaiKiran' in a 'MINGW64' environment at the directory '~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging' on the 'main' branch. The user enters '\$ git merge GitNewBranch'. The output shows the commit hash '962569a..8a2ea4a' and 'Fast-forward'. It then lists the changes: 'branch.txt => Week_6_GIT/Exercise_3_Branching_and_Merging/branch.txt | 0' and '1 file changed, 0 insertions(+), 0 deletions(-)'. Finally, it shows 'rename branch.txt => Week_6_GIT/Exercise_3_Branching_and_Merging/branch.txt (100%)'. The prompt '\$' is shown again at the bottom.

```
MINGW64:/c/Users/HP/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_...
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git merge GitNewBranch
Updating 962569a..8a2ea4a
Fast-forward
 branch.txt => Week_6_GIT/Exercise_3_Branching_and_Merging/branch.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
 rename branch.txt => Week_6_GIT/Exercise_3_Branching_and_Merging/branch.txt (100%)
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$
```

`git log --oneline --graph --decorate`


```

* 8a2ea4a (HEAD -> main, GitNewBranch) Add branch.txt in GitNewBranch in Ex3
* b53a8f7 Removed branch.txt from Root folder
* 962569a Add branch.txt in GitNewBranch
* a298904 Remove .gitignore
* c355346 (origin/main, origin/HEAD) Added PDF and comparison Screenshot
* abc4db5 Add screenshots of Exercise2
* 80bd781 Add .gitignore to ignore log files and folders
* d4eadd9 Added pdf for Browser compatible view without downloading the docx
* 735776b Added README.md for Week_6_GIT/Exercise_1
* 2be75d4 Ignore Microsoft Office temporary files
* f5c3215 Delete Week_6_GIT/Exercise_1/~$ercise_1.docx
* 21071a8 Completed Exercisel added document and screenshots
* 1931248 Completed Exercisel
* 051e890 Added README.md for Week_5_ReactJS_HOL/Exercise_17_Fetch_User_Data_from_REST_API
* 63b262d Added Week 5 Exercise 17 Fetch User Data from REAST API
* 23a3b8d Added README.md for Week_5_ReactJS_HOL/Exercise_16_Create_mailregisterapp/
* 25c7e3d Added Week 5 Exercise 16 Create Mail Register App
* aad9123 Added README.md for Week_5_ReactJS_HOL/Exercise_15_Create_ticketraisingapp
* 4bb799d Added Week 5 Exercise 15 Create ticket raising app
* 2a0614b Added README.md for Week_5_ReactJS_HOL/Exercise_14_React_Context_API
* ...skipping...
* 8a2ea4a (HEAD -> main, GitNewBranch) Add branch.txt in GitNewBranch in Ex3
* b53a8f7 Removed branch.txt from Root folder
* 962569a Add branch.txt in GitNewBranch
* a298904 Remove .gitignore
* c355346 (origin/main, origin/HEAD) Added PDF and comparison Screenshot
* abc4db5 Add screenshots of Exercise2
* 80bd781 Add .gitignore to ignore log files and folders
* d4eadd9 Added pdf for Browser compatible view without downloading the docx
* 735776b Added README.md for Week_6_GIT/Exercise_1
* 2be75d4 Ignore Microsoft Office temporary files
* f5c3215 Delete Week_6_GIT/Exercise_1/~$ercise_1.docx
* 21071a8 Completed Exercisel added document and screenshots
* 1931248 Completed Exercisel
* 051e890 Added README.md for Week_5_ReactJS_HOL/Exercise_17_Fetch_User_Data_from_REST_API
* 63b262d Added Week 5 Exercise 17 Fetch User Data from REAST API
* 23a3b8d Added README.md for Week_5_ReactJS_HOL/Exercise_16_Create_mailregisterapp/
* 25c7e3d Added Week 5 Exercise 16 Create Mail Register App
* aad9123 Added README.md for Week_5_ReactJS_HOL/Exercise_15_Create_ticketraisingapp
* 4bb799d Added Week 5 Exercise 15 Create ticket raising app
* 2a0614b Added README.md for Week_5_ReactJS_HOL/Exercise_14_React_Context_API
* 70cbe8a Added Week 5 Exercise 14 React Context API
* 23f33c3 Added README.md for Week_5_ReactJS_HOL/Exercise_13_React_Conditional_Rendering_Lists_and_Keys
* 544a05f Added Week 5 Exercise 13 React Conditional Rendering Lists and Keys
* 6403ae4 Added README.md for Week_5_ReactJS_HOL/Exercise_12_React_Conditional_Rendering/
* 6e73759 Added Week 5 Exercise 12 react Conditional rendering
* 2825923 Added README.md for Week_5_ReactJS_HOL/Exercise_11_React_Events
* d239516 Added Week 5 Exercise 11 React Events
* 58b30a0 Added README.md for Week_5_ReactJS_HOL/Exercise_10_React_JSX
* 5567087 Added Week 5 Exercise 10 React JSX
* dbd7822 Added README.md for Week_5_ReactJS_HOL/Exercise_9_ES6_Features_in_ReactJS
* 94e1c12 Added Week 5 Exercise 9 ES6 Features in ReactJS
* a892b2c Added README.md for Week_5_ReactJS_HOL/Exercise_8_React_State
* 8bfc7d6 Added Week 5 Exercise 8 React State
* 55bcb66 Added README.md for Week_5_ReactJS_HOL/Exercise_7_Props
* 64e6161 Added Week 5 Exercise 7 Props
* 2c86061 Added README.md for Week_5_ReactJS_HOL/Exercise_6_React_Router
* c2f56b1 Added Week 5 Exercise 6 React Router

```

Step 17: Delete the Branch

git branch -d GitNewBranch

Expected Output:

Deleted branch GitNewBranch (was abc1234).

Step 18: Verify Branch Deletion

git branch

Expected Output:

* main

Only the main branch remains.

Step 19: Verify Repository Status

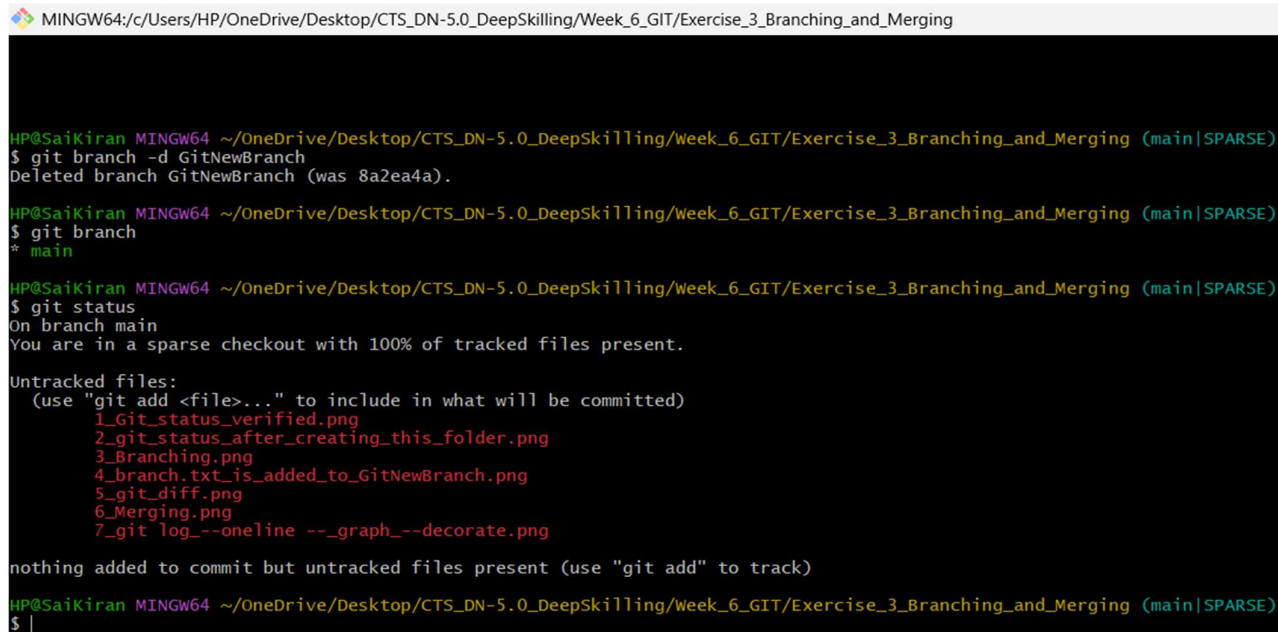
git status

Expected Output:

On branch main

nothing to commit, working tree clean

This confirms that the merge is complete and the repository is in a clean state.



The screenshot shows a terminal window with the following content:

```
MINGW64:/c:/Users/HP/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 8a2ea4a).

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git branch
* main

HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git status
On branch main
You are in a sparse checkout with 100% of tracked files present.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
       1_Git_status_verified.png
       2_git_status_after_creating_this_folder.png
       3_Branching.png
       4_branch.txt_is_added_to_GitNewBranch.png
       5_git_diff.png
       6_Merging.png
       7_git_log_--oneline --_graph_--decorate.png

nothing added to commit but untracked files present (use "git add" to track)
HP@Saikiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillig/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ |
```

Push Changes to GitHub

If you want the merged changes to be reflected on GitHub:

git pull origin main

git push origin main


```
HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git pull origin main --rebase
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 3.78 KiB | 94.00 KiB/s, done.
From https://github.com/G-Manikanta1729/CTS_DN-5.0_DeepSkillling
* branch      main      -> FETCH_HEAD
   c355346..81256b4 main    -> origin/main
Successfully rebased and updated refs/heads/main.

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ git push origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 12 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 1.05 KiB | 539.00 KiB/s, done.
Total 12 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 2 local objects.
To https://github.com/G-Manikanta1729/CTS_DN-5.0_DeepSkillling.git
   81256b4..57916d2  main -> main

HP@SaiKiran MINGW64 ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling/Week_6_GIT/Exercise_3_Branching_and_Merging (main|SPARSE)
$ |
```

Final Project Structure

CTS_DN-5.0_DeepSkillling/

|

├── Week_1_Engineering_Concepts/

├── Week_2_Spring_Framework/

├── Week_3_Spring_REST/

├── Week_4_Microservices/

├── Week_5_ReactJS_HOL/

└── Week_6_GIT/

├── Exercise_1_Git_Basics/

├── Exercise_2_Git_Ignore/

└── Exercise_3_Branching_and_Merging/

└── branch.txt

Command Summary

cd ~/OneDrive/Desktop/CTS_DN-5.0_DeepSkillling

git status

git branch GitNewBranch

```
git branch
```

```
git checkout GitNewBranch
```

```
echo "This file is created in GitNewBranch." > branch.txt
```

```
git add branch.txt
```

```
git commit -m "Add branch.txt in GitNewBranch"
```

```
git status
```

```
git checkout main
```

```
git diff main GitNewBranch
```

```
git difftool main GitNewBranch # Optional (requires P4Merge)
```

```
git merge GitNewBranch
```

```
git log --oneline --graph --decorate
```

```
git branch -d GitNewBranch
```

```
git status
```

```
git pull origin main
```

```
git push origin main
```

This implementation follows the sequence described in the CTS Hands-on document: create a branch, switch to it, make changes, commit them, compare branches, merge into the main branch, view the commit log, and delete the branch.

Exercise_3_Branching_and_Me

Start backup > ... Week_6_GIT > Exercise_3_Branching_and_Merging

Search Exercise_3_Branching_anc

New

Copy Paste Print Share Delete

Sort View

Details

Home

Gallery

Desktop

Downloads

Documents

Pictures

Music

Videos

Exercise_1

Screenshots

Exercise_2_Git_Ic

Exercise_3_Branc

1_Git_status_verifi
ed

2_git_status_after
_creating_this_fol
der

3_Branching

4_branch.txt_is_a
dded_to_GitNew
Branch

5_git_diff

6_Merging

7_git
log_--oneline
--graph--decor
ate

8_Deletion_of_br
anch

branch

9 items